

RCIM Model-Bank Reproduction MLP Family Full-Matrix Repair Campaign Results Report

Feedforward Transmission Error Baseline

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Overview

This report closes the dedicated exact-paper MLP family repair wave prepared in:

- `doc/reports/campaign_plans/track1/exact_paper/2026-04-21-17-20-12_track1_mlp_family_full_matrix_repair_campaign_plan_report.md`

The campaign targeted only the MLP family-target pairs selected for the post-relaunch repair pass across the canonical RCIM Model-Bank Reproduction full-matrix surfaces.

- campaign name: `track1_mlp_family_full_matrix_repair_campaign_2026_04_21_17_20_12`
- report timestamp: `2026-04-21-22-19-09`
- completed validation runs: `324`
- failed validation runs: `0`
- promoted targeted pairs: `1/12`
- retained baseline pairs: `11/12`
- visibly promoted benchmark cells: `1/24`
- campaign artifact root:
`output/training_campaigns/track1/exact_paper/forward/family_repair/mlp/track1_mlp_family_full_matrix_repair_campaign_2026_04_21_17_20_12`

Campaign Winner

- Run: `track1_mlp_amplitude_0_closure_attempt_17`
- Scope: `amplitudes_only`
- Closure Score: `1.000`
- Met / Near / Open: `2 / 0 / 0`

Targeted Pair Outcome

The table below compares the accepted pre-wave baseline against the best campaign-local candidate and the final accepted post-wave value for each targeted MLP pair. Each metric cell is shown as MAE / RMSE .

Pair	Baseline	Campaign Best	Accepted	Source	Result
A0	0.0093971 / 0.013702	0.0093971 / 0.013702	0.0093971 / 0.013702	baseline	fully_green
A1	0.00755033 / 0.0137694	0.00755033 / 0.0137694	0.00755033 / 0.0137694	baseline	still_non_green
A3	0.00552258 / 0.00892685	0.00552258 / 0.00892685	0.00552258 / 0.00892685	baseline	fully_green
A39	0.00520776 / 0.00919113	0.00520776 / 0.00919113	0.00520776 / 0.00919113	baseline	fully_green
A40	0.00699309 / 0.0120674	0.00672959 / 0.0128006	0.00672959 / 0.0120674	mixed	fully_green
A81	0.00536461 / 0.00939595	0.00536461 / 0.00939595	0.00536461 / 0.00939595	baseline	fully_green
A156	0.00682945 / 0.011491	0.00682945 / 0.011491	0.00682945 / 0.011491	baseline	still_non_green
A240	0.00638307 / 0.0121452	0.00638307 / 0.0121452	0.00638307 / 0.0121452	baseline	still_non_green
phi1	0.00582546 / 0.00916257	0.00582546 / 0.00916257	0.00582546 / 0.00916257	baseline	fully_green
phi3	0.0590324 / 0.0742506	0.0590324 / 0.0742506	0.0590324 / 0.0742506	baseline	fully_green
phi39	0.0531587 / 0.0696308	0.0531587 / 0.0696308	0.0531587 / 0.0696308	baseline	fully_green
phi162	0.826864 / 1.33334	0.826864 / 1.33334	0.826864 / 1.33334	baseline	still_non_green

Accepted MLP Row After Closeout

- Table 2 remaining non-green harmonics: 1, 156, 240
- Table 3 remaining non-green harmonics: 1, 240
- Table 4 remaining non-green harmonics: 162
- Table 5 remaining non-green harmonics: 162
- total remaining non-green MLP cells on Tables 2-5 : 7

Canonical RCIM Model-Bank Reproduction Impact

- This MLP -only wave updates the accepted MLP family row without reopening the already closed cross-family open-cell closeout.
- The benchmark row now keeps the better visible metric cell between the accepted baseline and this dedicated repair wave.
- Global RCIM Model-Bank Reproduction cross-family closure counts remain unchanged at 7/10 , 10/10 , 9/9 , and 9/9 across Tables 2-5 .

Resulting Canonical State

- benchmark path: `doc/reports/analysis/rcim_paper_reference/RCIM Paper Reference Benchmark.md`
- master summary path: `doc/reports/analysis/Training Results Master Summary.md`
- final closeout report path: `doc/reports/campaign_results/track_1/exact_paper/forward/2026-04-21-22-19-09_track1_mlp_family_full_matrix_repair_campaign_results_report.md`